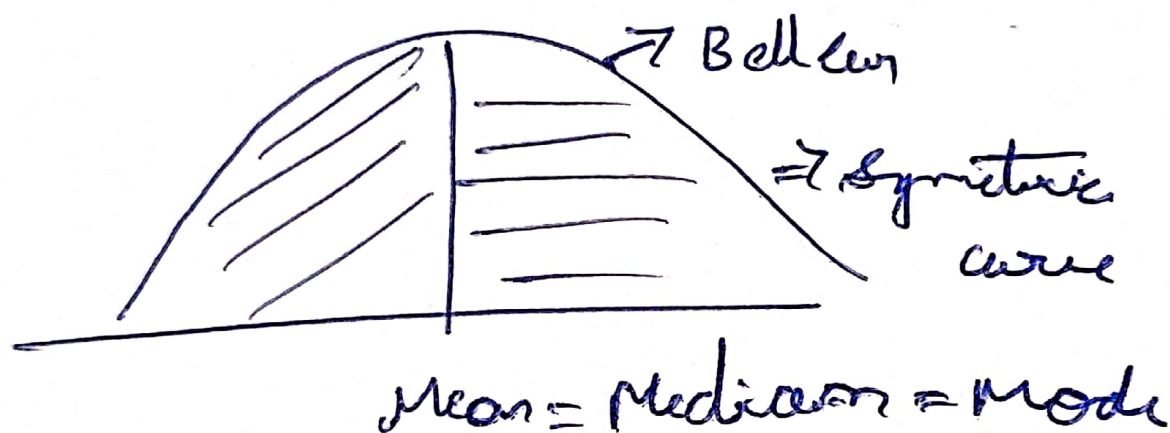


Normal or Gaussian distribution 6th

In probability and Statistics a normal distribution or Gaussian distribution is a type of continuous probability distribution for a real-valued Random Variable

①



Notation $N(\mu, \sigma^2)$

Parameters :- $\mu \in \mathbb{R} = \text{mean}$

$\sigma^2 \in \mathbb{R} > 0 = \text{variance}$
 $x \in \mathbb{R}$

Eg : - weight of student in a class
height of students in a class

IRIS Dataset

$$P.D.F = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2}$$

Mean

$$\mu = \sum_{i=1}^n \frac{x_i}{n}$$

Variance

$$\sigma^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{n}$$

$$\sigma = \sqrt{\text{Variance}}$$

Empirical rule of.

68-95-99.7

Problem

$$P(\mu - \sigma \leq x \leq \mu + \sigma) \approx 68\%$$

$$P(\mu - 2\sigma \leq x \leq \mu + 2\sigma) \approx 95\%$$

$$P(\mu - 3\sigma \leq x \leq \mu + 3\sigma) \approx 99.7\%$$